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Wenting Zhao, Xiang Ren, Jack Hessel, Claire Cardie, Yejin Choi, and Yuntian Deng. 2024. Wildchat: 1m chatgpt interaction logs in the wild. *Preprint*, arXiv:2405.01470.

A Dataset Creation

To create the dataset, we first split the examples into knowledge-based examples, following a method similar to Simhi et al. (2024). See illustration in Figure 6.

Specifically, we performed one greedy generation and five generations with a temperature of 0.5. We used a 3-shot in-context learning scenario, generating a maximum of 5 tokens, and considered a generation correct only if it exactly matched the factually correct answer.

We also adopted the basic dataset curation process described in Simhi et al. (2024), but with two key modifications: we started with 70K examples instead of 30K, and we generated 10 tokens instead of 5. Each example in the dataset begins with `question:` and ends with `answer:`.

For instruct models, we adjusted the format to align better with their structure. Specifically, we presented the few-shot examples as a user-assistant conversation, where the user asks the questions and the assistant provides the answers. Additionally, we replaced `answer:` with `The answer is`, as part of the assistant generation, since this change was observed to improve the performance of instruction models.

To split the knowledge examples into factually correct examples and hallucination-despite knolwedge examples, we sampled 20K knowledge-based examples (or fewer if fewer were available). Using the prompt settings, we checked whether the generated text changed and whether the exact match for the correct answer appeared within the 10-token model generations.

A.1 Additional Refinement

We observed certain issues, especially with the instruct models, where an exact match was insufficient. For example, the model sometimes failed to generate an answer or produced a correct answer with minor variations, such as synonyms. To address these issues, we curated the WACK examples further, applying a set of simple heuristics that proved effective during manual examination:

1. **Removing negations:** We excluded examples where the generation stated with "The answer is not."

2. **Synonyms:** Using the NLTK library (Loper and Bird, 2002), we removed examples where a synonym of the correct answer appeared in the generated text.

3. **Stem-based similarity:** We excluded examples if the stemmed version of the generation and the factually correct answer shared more than half of their words.

4. **Edit distance:** We kept examples where the edit distance between the generated text and the correct answer (in their stemmed versions) was greater than 2, or the answer is a number and `great`, `none`, and `n/a`, which were removed if present in the generated answer.

5. **Initial word match:** We removed examples where the generated answer was the first word of the factually correct answer.

6. **Special formatting:** For GEMMA-INSTRUCT model, which we saw that typically generates the final answer enclosed in `**`, we removed examples where this formatting was absent.

Thus, we removed between 10% and 45% of all the hallucination examples. Note that this is a very harsh criterion for removing hallucinations; however, since our aim was to demonstrate that certain hallucinations exist, we preferred to remove any possibility of wrongly classified hallucinations.

A.2 Additional Implementation Details

We use the datasets under the Apache License and the models under each one's agreement terms to follow each artifact's terms. All experiments were run on NVIDIA RTX 6000 Ada (49GB) with 4 CPUs. Generating all the datasets and results takes approximately one month on one GPU. For probe training we used Sklearn (Pedregosa et al., 2011) Logistic regression running 1000 iterations.

Lastly, We used AI assistants only for simple paraphrasing as part of writing this paper.

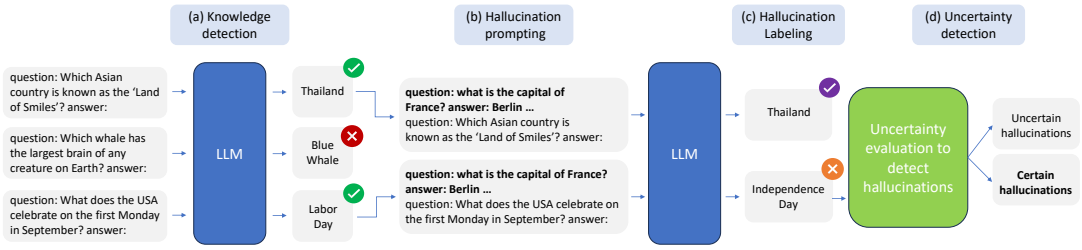

Figure 6: **Overview of finding CHOKE**: This is an extended figure from (Simhi et al., 2024) with approval, including the certainty step. The first step is evaluating the model's knowledge; next, we use a hallucination-inducing prompt to elicit hallucination despite encoded knowledge. Lastly, we take those hallucination examples and evaluate their certainty to detect CHOKE examples.

B Qualitative Evaluation

In this section, we show qualitative examples of certain hallucinations. In Table 3 provides an example from each model using Prompt 4 on the Natural Questions dataset. These are examples of CHOKE hallucinations, where the model outputs have high probability and low semantic entropy, indicating a high degree of certainty.

C Certainty Methods Additional Specifics

In this section, we elaborate on specifics in the calculations of the different methods.

C.1 Probability and Probability Difference

To ensure that the probability we consider corresponds to the probability of the actual answer and not a preceding token, we employed the following heuristic: skipping over any of the following tokens: `"<|assistant|>"`, `"<|user|>"`, `"<|begin_of_text|>"`, `"<|end_of_text|>"`, `"<|eot_id|>"`, `"<|start|>"`, `"<|end|>"`, `"<|sep|>"`, `"<|sep_id|>"`, `"assistant"`, `"user"`, `"\n"`, `"answer"`, `"The"`, `"Answer"`, `"¨"`, `"'"`, `" answer"`, `"is"`, `"it"`, `"it's"`, `":"`, `" "`, `" is"`, `" correct"`, `"correct"`, `"*"`, `"**"`, `" **"`.

This heuristic proved sufficient during a manual investigation.

C.2 Sampling Based Methods

For the Semantic Entropy, Sampling, and Predictive Entropy methods, it is necessary to consider the temperature and define a stopping condition for each generation.

We stopped the generation if one of the following sequences was produced: '\n\n\n\n', '\n\n\', "\n\n, 'Question:', 'Context:', ".\n", ". ", 'question:', "Alice", "Bob", "(", "Explanation", "\n question:", "What", "\n answer". These sequences often indicate the generation of new text that is not relevant to answering the question.

We used a temperature of 1 for 10 generations and an additional generation with a low temperature of 0.1, following the approach in the code of Kuhn et al. (2023) in repository `https://github.com/jlko/semantic_uncertainty`, and using DeBERTa (He et al., 2020) as the entailment model for the clustering stage based on meaning similarity. Additional results with a temperature of 0.5 instead of 1 are presented in Appendix E. Note that we used 10 tokens to generate as the maximum to be consistent with the knowledge and dataset creation steps.

C.3 Mitigation Metrics

In Section 5 we evaluate the mitigation abilities of sampling and predictive entropy. In this section we detail each metric.

Sampling. Sampling-based methods assess the diversity of the model's generated outputs, under the assumption that greater diversity reflects lower certainty. Following Cole et al. (2023), we define diversity as the proportion of unique outputs in S generated samples. The uncertainty score is calculated as $1 - |U|/|S|$, where U is the set of unique generations, and $|U|/|S|$ represents the ratio of unique outputs to the total number of generations.

Predictive Entropy. Predictive Entropy estimates uncertainty by evaluating the average unpredictability of the model's outputs. We approximate predictive entropy following Kuhn et al. (2023) and Tomani et al. (2024) by estimating the uncertainty

Model	Prompt	Response		Uncertainty metric	
		Original Response	Hallucinated Response	Probability	Semantic Entropy
Gemma	question: what is the measure of the number of different species present in an area?	biodiversity	species richness	0.31	0.0
Llama	question: who wrote the song it's the climb?	alexander	Miley Cyrus	0.42	$1.11e^{-}16$
Mistral	question: if there is a random change in the genetics of a small population it is termed?	genetic drift	mutation	0.49	0.14
Gemma-Instruct	question: who played the mom on lost in space?	June Lockhart	Molly Parker	0.89	0.23
Llama-Instruct	question: what gas is given off by concentrated hydrochloric acid?	hydrogen chloride	hydrogen gas (H_2)	0.98	$2.22e^{-}16$
Mistral-Instruct	question: who published harry potter and the prisoner of azkaban?	Scholastic Inc	J.K. Rowling	0.99	$2.22e^{-}16$

Table 3: CHOKE generated answers and uncertainty measures were obtained using greedy decoding on the Natural Questions dataset in Prompt 4. In each of these examples, the model generates a hallucination despite having the necessary knowledge. The probability of the generated answer is high, and the semantic entropy is low, indicating that these examples exhibit high certainty.

of the model based on its generations for a given prompt x. Using L generated samples, the predictive entropy is calculated as:

$$PE \approx -1/L \sum_{i=1}^{L} logp(l_i|x) \quad (5)$$

Here, $p(l_i|x)$ represents the likelihood of the i-th generation given the prompt x. Predictive entropy captures the average uncertainty across the generated outputs.

We investigate whether these uncertainty measures can reliably detect and mitigate CHOKE hallucinations.

D Certain HK+ Exist – Additional Results

In this section, we present results similar to those in Section 4, First we show the main result on TriviaQA in Table 4, where we can see the existence of CHOKE also across permutations on TriviaQA.

Next, focusing on the Gemma and Llama models. See Figures 8 and 7. These results correlate with those in the main paper on the Mistral model and demonstrate that CHOKE hallucinations exist across thresholds. Furthermore, they show that these hallucinations occur across different methods and that instruct-tuned models exhibit poorer calibration between certainty and hallucinations.

Lastly, In Figure 9 we show the existences of CHOKE on all our seven different prompts on Mistral Natural Question setting.

E Semantic Entropy Results – Different Temperature

In Section 4.3, we used Semantic Entropy with a temperature of 1 for generating the samples. To demonstrate that the certainty results are not specific to this temperature, we present in Figure 10 the Semantic Entropy results on the Mistral models. In the left subfigure, we show the results using a temperature of 1, and in the right, we show the results using a temperature of 0.5. We observe that under a temperature of 0.5, there are even more certain hallucinations, further proving that the certainty hallucination phenomenon is not specific to a temperature of 1.

F Prompt Selection

As described in Section 3.1, we use 7 variants of neutral prompts for hallucination inducing, as can be seen in Table 5. To strengthen the credibility